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Practical assignment 4

CMPG 313

Conversion of FOL rules to English:

1. **Disease(flu) Diagnosis Rule**:
   There is someone X has flu if X has a fever, cough, and sore throat.
2. **Disease(common) Cold Diagnosis Rule**:
   There is a person X has a common cold if X has sneezing, runny nose, and mild fever.
3. **Disease(malaria) Diagnosis Rule**:
   There is a person X has malaria if X has fever, chills, sweating, and headache.
4. **Disease(covid-19) Diagnosis Rule**:
   There is a person X has COVID-19 if X has fever, cough, shortness of breath, and loss of taste.
5. **Disease (strep throat) Diagnosis Rule**:
   A person which is X has strep throat if X has sore throat, swollen lymph nodes, and fever.

Define an advice base:

For Flu:

-Get enough rest.

-Make sure that you are always hydrated by drinking fluids.

-Always drink your medication.

For Common cold:

-Use nasal spray or drops.

-Rinse with warm water that has salt for sore throat.

-Get enough rest and drink lots of fluids.

For Malaria:

-Immediately get medical help.

-Drink all your prescribed antimalarial medications.

-Maintain your fluid intake.

For covid-19:

-Isolate yourself as soon as possible.

-Monitor your breathing levels.

-Look for medical help if you experience shortness of breathing.

For strep throat:

-Go and see the Doctor for antibiotics.

-Rinse with warm salt water.

-Get enough rest and drink warm liquids.

Debug and fix code errors:

```
def healthcare_chatbot():
    # Knowledge base: diseases and their symptoms
    knowledge_base = {
        'flu': ['fever', 'cough', 'sore_throat'],
        'common_cold': ['sneezing', 'runny_nose', 'mild_fever'],
        'malaria': ['fever', 'chills', 'sweating', 'headache'],
        'covid19': ['fever', 'cough', 'shortness_of_breath', 'loss_of_taste'],
        'strep_throat': ['sore_throat', 'swollen_lymph_nodes', 'fever']
    }

    # Advice base
    advice_base = {
        'flu': [
            "Get plenty of rest",
            "Stay hydrated by drinking fluids",
            "Consider taking over-the-counter fever reducers like acetaminophen"
        ],
        'common_cold': [
            "Use saline nasal drops or spray",
            "Gargle with warm salt water for sore throat",
            "Get extra rest and drink plenty of fluids"
        ],
```

```
    'malaria': [
        "Seek immediate medical attention",
        "Take prescribed antimalarial medications",
        "Rest and maintain fluid intake"
    ],
    'covid19': [
        "Self-isolate immediately",
        "Monitor your oxygen levels with a pulse oximeter",
        "Seek medical care if you have difficulty breathing"
    ],
    'strep_throat': [
        "See a doctor for antibiotics",
        "Gargle with warm salt water",
        "Get plenty of rest and drink warm liquids"
    ]
}

print("Welcome to the Healthcare Chatbot!")
print("Please answer the following questions about your symptoms (yes/no):")

# Collect symptoms from user
user_symptoms = []
all_symptoms = set()
for symptoms in knowledge_base.values():
    all_symptoms.update(symptoms)

for symptom in sorted(all_symptoms):
    response = input(f"Do you have {symptom.replace('_', ' ')}? ").lower()
    if response == 'yes':
```

```
            user_symptoms.append(symptom)

    # Diagnose based on symptoms
    possible_diseases = []
    for disease, symptoms in knowledge_base.items():
        if all(symptom in user_symptoms for symptom in symptoms):
            possible_diseases.append(disease)

    # Provide diagnosis and advice
    if possible_diseases:
        print("\nBased on your symptoms, you might have:")
        for disease in possible_diseases:
            print(f"- {disease.replace('_', ' ').title()}")
            print("Recommended actions:")
            for advice in advice_base[disease]:
                print(f"  * {advice}")
            print()
    else:
        print("\nNo specific diagnosis could be made based on your symptoms.")
        print("Please consult a healthcare professional for further evaluation.")

# Run the chatbot
if __name__ == "__main__":
    healthcare_chatbot()
```

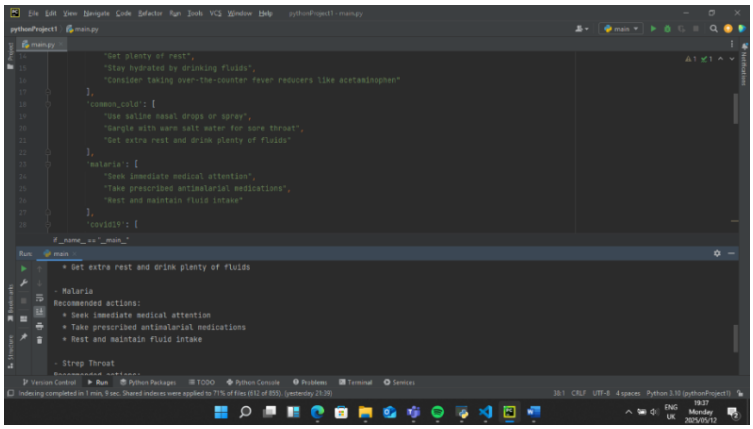
Report:

Introduction

The report is for the development and testing of a First-Order Logic based Healthcare Chatbot designed to diagnose diseases based on the symptoms reported by the user. This system uses a knowledge base of symptom-disease rules and an advice base to give recommendations.

Screenshot Test Cases:

Below are the test cases for all the diseases with our Chatbot diagnosis and their output.


pythonProject1 - main.py
"Get plenty of rest",
"Stay hydrated by drinking fluids",
"Consider taking over-the-counter fever reducers like acetaminophen"
],
'common_cold': [
"Use saline nasal drops or spray",
"Gargle with warm salt water for sore throat",
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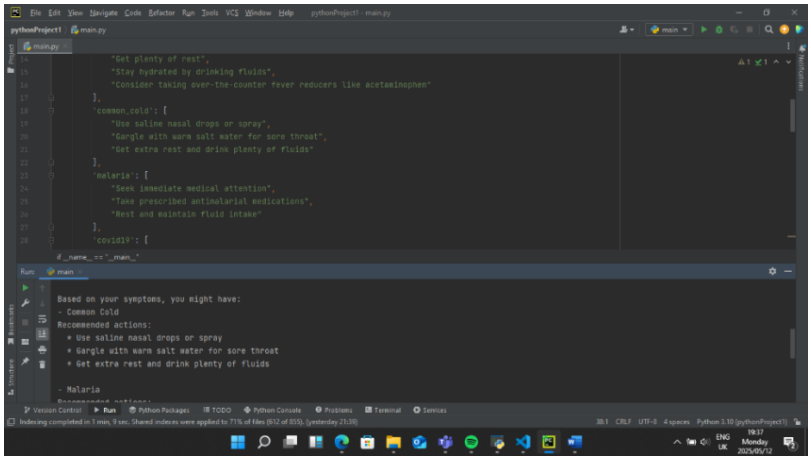
],
'malaria': [
"Seek immediate medical attention",
"Take prescribed antimalarial medications",
"Rest and maintain fluid intake"
],
'covid19': [
if __name__ == "__main__"
* Get extra rest and drink plenty of fluids
- Malaria
Recommended actions:
* Seek immediate medical attention
* Take prescribed antimalarial medications
* Rest and maintain fluid intake
- Strep Throat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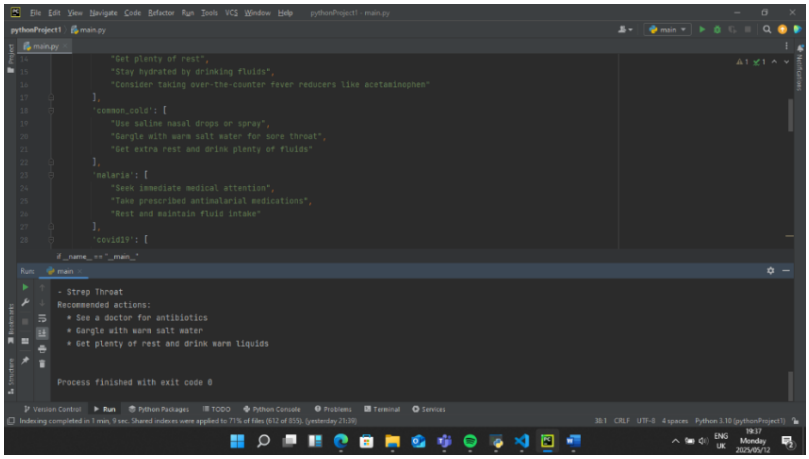
"Take prescribed antimalarial medications",
"Rest and maintain fluid intake"
],
'covid19': [
if __name__ == "__main__"
Based on your symptoms, you might have:
- Common Cold
Recommended actions:
* Use saline nasal drops or spray
* Gargle with warm salt water for sore throat
* Get extra rest and drink plenty of fluids
- Malaria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"Take prescribed antimalarial medications",
"Rest and maintain fluid intake"
],
'covid19': [
if __name__ == "__main__"
- Strep Throat
Recommended actions:
* See a doctor for antibiotics
* Gargle with warm salt water
* Get plenty of rest and drink warm liquids
Process finished with exit code 0

Proposed Enhancements

To improve the chatbot the following features can be added.

Feature 1: Severity Assessment

-Description: Organize disease severity (for example mild/ moderate/severe) based on symptom severity.

-Implementation:

-ask follow-up questions

-use levels to determine urgency.

Feature 2: Multilingual Support

-Description: Allow users to interact using different languages.

-Implementation:

-Incorporate a translation in multiple languages.

Feature 3: Symptom Tracker and History

-Description: Track the users' symptoms from time to time for trend analysis.

-Implementation:

-create simple visualizations.

Conclusion:

I can say that the chatbot successfully diagnosed diseases using FOL and it offers actionable advice and for future purposes it will include implementing the proposed enhancements above to improve usability and accuracy.